

# Edit Smali Code – Complete Guide

---

Smali code edit karna APK modding ka sab se important aur powerful step hai. JADX-GUI se aap code logic samajhte hain, aur APKTool se nikali gayi smali files mein aap actual modifications karte hain. Smali Android ke Dalvik bytecode ki human-readable form hai, aur isko edit karke aap app ke behavior ko change kar sakte hain.

## Smali Code Kya Hai?

Smali assembly-level language hai jo DEX bytecode ko human-readable format mein represent karti hai. Jab aap `apktool d app.apk` run karte hain, toh har `.class` file smali format mein convert ho jati hai. Har class ki apni smali file hoti hai jismein uske methods, variables, aur logic ka low-level representation hota hai.

## Smali Code Structure

Smali code ka basic structure kuch is tarah hota hai:

**.class directive** – Yeh class ka name aur access modifiers define karta hai. Example: `.class public Lcom/example/app/MainActivity;`

**.super directive** – Yeh parent class specify karta hai.

Example: `.super Landroid/app/Activity;`

**.method directive** – Yeh method define karta hai. Method ka naam, parameters, aur return type ismein specify hote hain. Example: `.method public  
onCreate(Landroid/os/Bundle;)V`

**.locals directive** – Yeh method mein local variables ki total number batata hai. Example: `.locals 3`

**.registers directive** – Yeh total registers (local + parameters) ki number batata hai. Example: `.registers 5`

**Parameter Registers** – `p0` this reference (object itself) hota hai, `p1` pehla parameter, `p2` doosra parameter, aur isi tarah.

**Local Registers** – `v0`, `v1`, `v2`, etc. local variables store karte hain.

**Return Types** – `V` void (kuch return nahi), `Z` boolean, `I` integer, `J` long, `F` float, `D` double, `Ljava/lang/String;` String object, `L<package/Class>;` custom object.

## Common Smali Operations

### Move Operations (Value Assign Karna)

Move operations registers ke beech values transfer karte hain. `const/4 v0, 0x1` se `v0` mein 1 store hota hai. `const/16 v1, 0x64` se `v1` mein 100 store hota hai. `const-string v2, "Hello"` se `v2` mein string store hoti hai. `move`

v0, v1 se v1 ki value v0 mein copy hoti hai. move-result v0 se previous method call ka result v0 mein store hota hai.

## **Return Operations (Values Return Karna)**

return-void se void return hota hai. return v0 se v0 ki value return hoti hai.

## **If Conditions**

Smali mein conditions compare karne ke liye yeh instructions use hoti hain: if-eq v0, v1, :label (v0 == v1 then jump), if-ne v0, v1, :label (v0 != v1 then jump), if-lt v0, v1, :label (v0 < v1 then jump), if-le v0, v1, :label (v0 <= v1 then jump), if-gt v0, v1, :label (v0 > v1 then jump), if-ge v0, v1, :label (v0 >= v1 then jump). if-eqz v0, :cond\_0 se agar v0 == 0 hai toh cond\_0 par jump hota hai.

## **Method Calls**

invoke-virtual {v0, v1}, Lcom/example/MyClass;->myMethod(Ljava/lang/String;)V se v0 (object) aur v1 (parameter) ke saath method call hoti hai. move-result v1 se method ka result v1 mein store hota hai. invoke-static {v0}, Lcom/example/Utils;->helper(Ljava/lang/String;)V se static method call hoti hai.

## **New Object Creation**

new-instance v0, Lcom/example/MyClass se object create hota hai aur invoke-direct {v0},

Lcom/example/MyClass;-><init>()V se constructor call hota hai.

## **Array Operations**

const/4 v0, 0x5 aur new-array v1, v0, [I se 5 integers ka array create hota hai. aget v2, v1, v0 se array v1 ke index v0 ki value v2 mein store hoti hai. aput v2, v1, v0 se v2 ki value array v1 ke index v0 par store hoti hai.

## **Practical Examples**

### **Example 1: Always Return True**

Original Java code: `public boolean isProUser() {  
return checkLicense(); }`

Original Smali:

```
.method public isProUser()Z  
    .locals 1  
    invoke-direct {p0}, Lcom/example/App;-  
>checkLicense()Z  
    move-result v0  
    return v0  
.end method
```

Modified Smali (Always return true):

```
.method public isProUser()Z  
    .locals 1
```

```
    const/4 v0, 0x1
    return v0
.end method
```

## **Example 2: Skip License Check**

Original Smali mein conditional check hoti hai. Modified Smali mein goto :cond\_0 laga kar directly premium content par jump kar sakte hain.

## **Example 3: Toast Message Remove Karna**

Original Smali mein invoke-virtual {v0}, Landroid/widget/Toast;->show()V call hoti hai. Is line ko comment out karne se toast remove ho jati hai.

## **Practical Workflow**

### **Step 1: APK Decompile Karein**

```
apktool d app.apk -o decompiled/
```

**Step 2: Target Smali File Find Karein** JADX-GUI se class aur method ka name note karein.

decompiled/smali/com/example/app/MainActivity.smali mein jayein.

**Step 3: Method Locate Karein** .method public methodName search karein. Conditional checks, return values, method calls identify karein.

**Step 4: Edit Karein** Text editor (VSCode, Notepad++, vim) use karein. Modify logic according to requirements.

### **Step 5: APK Rebuild Karein**

```
apktool b decompiled/ -o modified.apk
```

### **Step 6: Sign APK Karein**

```
jarsigner -verbose -sigalg SHA1withRSA -digestalg  
SHA1 -keystore my.keystore modified.apk alias
```

## **Smali Tips**

**JADX-GUI Reference Use Karein:** JADX-GUI mein code padhein, understand karein, phir smali mein modify karein.

**Small Changes Start Karein:** Pehle chhoti cheezein change karein (strings, constants) aur test karein.

**Original Code Backup:** Modify karne se pehle original smali file ka backup zaroor lein.

**Comments Use Karein:** Apni modifications ko comment karein taake later samajh aaye: # MOD: Always return true

**GOTO is Powerful for Skipping:** Goto instructions use karke aap code ke parts skip kar sakte hain.

**APK Rebuild Warnings:** Rebuild warnings normal hain, sign karte waqt ignore ho jati hain.

**Smali Syntax:** Hamesha correct syntax maintain karein, ek bhi syntax error se APK install nahi hogi.

## Common Mods

- **Remove License Check** – goto ke saath conditional skip karein
- **Always Return True** – const/4 v0, 0x1 and return v0 use karein
- **Remove Toast/Ads** – Method calls ko comment out karein
- **Change Strings** – const-string v0, "New Text" use karein
- **Add Logging** – invoke-static se log function call karein

 **Educational Purpose Only:** Yeh guide sirf educational aur learning purposes ke liye hai. Smali code editing ka istemal sirf apne own apps aur authorized environments mein karein. Kisi doosre ke app ko bina permission ke modify ya crack karna applicable cyber laws ke under punishable hai.